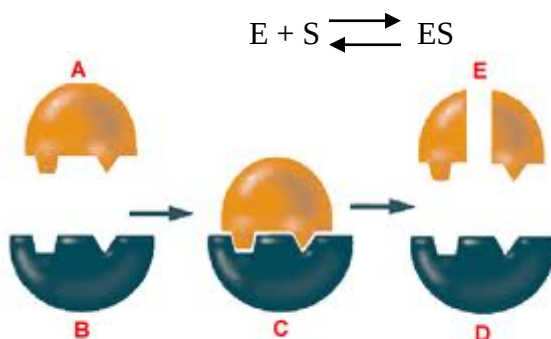


Lecture 20

Enzyme substrate complexes

Enzyme substrate complex

When an enzyme unites with its substrate, the resulting union is called enzyme substrate complex.



Enzyme Active site

The active site of an enzyme is a particular location of that enzyme that binds to the substrate and where catalysis occurs. It is constituted with mainly residues of the enzyme and in few cases contains a co-factor. Active site of a particular enzyme possesses a unique geometric shape and chemical properties that allow the enzyme to recognize a specific substrate and bind with it then.

Important properties of active site residues

- Charge (partial, dipoles, helix dipole)
- pKa
- Hydrophobicity
- Flexibility
- Reactivity

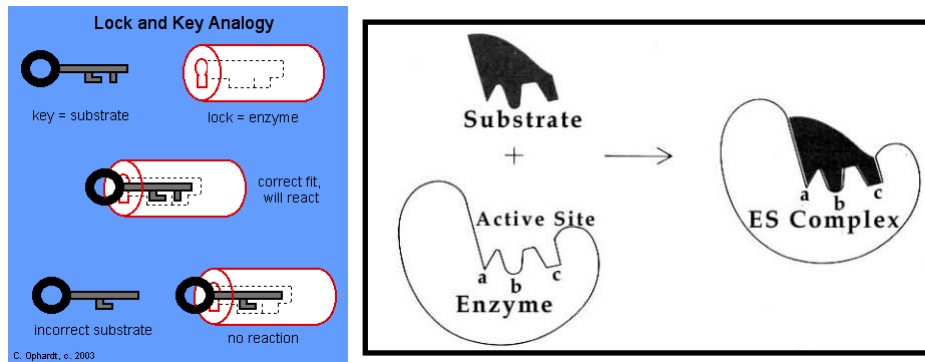
Substrate binding specificity

One of the most important properties of Enzymes are highly specific with respect to the identity of a substrate. Even, if in a reaction mixture, more than one substrate are present, the enzyme will bind to that substrate only which is complement with the active site with respect to geometric shape, electronic and stereospecificity.

There are two different models to describe how substrate binds with enzyme.

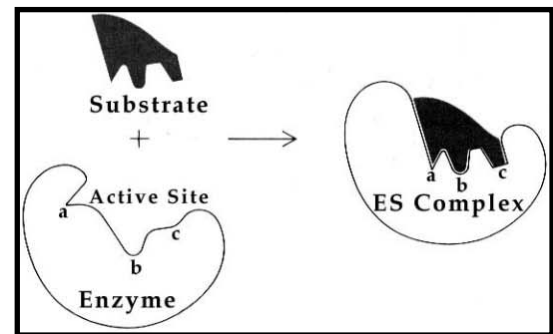
Lock and key model

The reaction specificity of an enzyme can be explained with the help of **lock and key** model where enzyme has analogy with lock, active site with key-hole and substrate with key. In case of lock and key, only correctly sized key has permit to open the lock. Similarly, only a substrate which has perfect sized matching with the active site of the enzyme can bind with the enzyme and subsequently reaction can occur.



Induced fit model

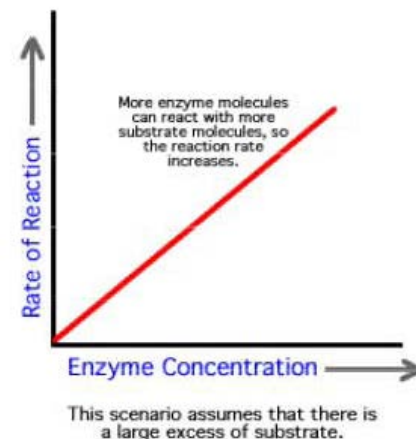
The rigid shaped binding of substrate and enzyme, assumed in the 'lock and key model' can not explain adequately all experimental fact. To find proper explanation induced fit model was proposed. In this model it was assumed that before binding, substrate does not fit exactly with the active site. But after binding, the substrate induce the active site of a structurally flexible enzyme in such a manner that it can fit with it. Thus, with the help of 'induced fit model' it can be explain how a single enzyme is capable of binding many substrate.



Factors affecting enzyme activity

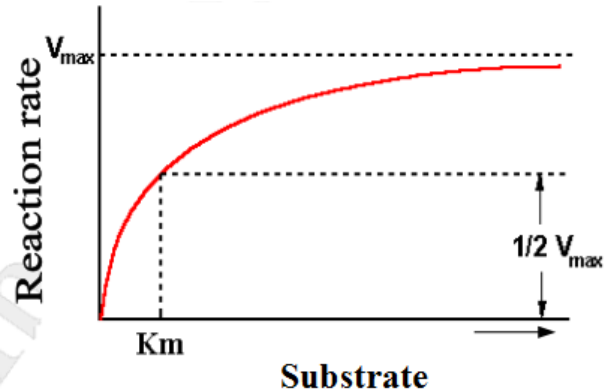
Enzyme concentration

Two enzyme molecules bind independently bind with two substrate molecule to give two enzyme substrate complexes whereas one enzyme molecule gives only one. Thus, the rate of an enzymatic reaction increases linearly with increasing enzyme concentration. Deviation from linear relation could be occur



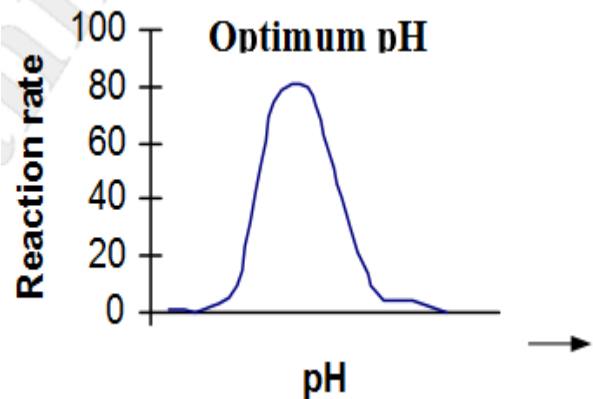
Substrate concentration

The enzyme catalyzed reaction rate increases as the substrate concentration increases until a certain point called $V_{\max}$ is reached where the reaction attains maximal velocity. Rate of the reaction does not exceed further on increase in substrate concentration because at $V_{\max}$ enzyme molecules are completely saturated with substrate molecules. At relatively low substrate concentration the reaction rate increases linearly with increasing substrate concentration. In this case the reaction follows first order kinetics. With further increase in substrate concentration the plot becomes curved where rate increase is not as much as in the low substrate concentration. At moderate substrate concentration the reaction follows mixed order kinetics. Once the maximum velocity ($V_{\max}$) point is reached after that with increase in temperature no more velocity is increased. In this case the reaction follows zero order kinetics.



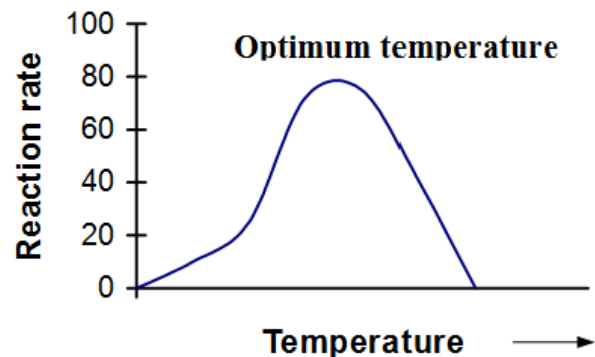
Effect of pH

A very little range of pH is effective for an enzyme to be active. Almost for all enzymes there is an optimum temperature where it shows maximum efficiency. The state of optimum condition may be arises due to the following reason: (a) a true reversible effect on its velocity itself, (b) an effect of pH on the affinity of enzyme for the substrate. (c) the effect of pH on the stability of enzyme. Denaturation of the enzyme may takes place on either side of the optimum pH value. All these effects may operate simultaneously.



Effect of temperature

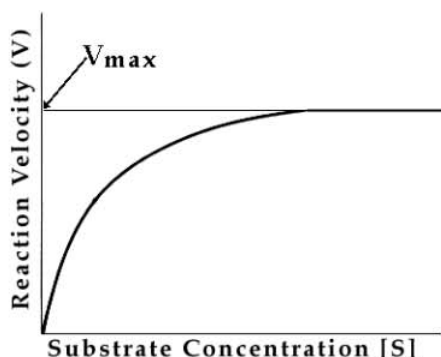
At very low temperature enzyme does not shows its activity. The rate of an enzymatic reaction increases with increase of temperature until it reaches to the maximum. Further increase of temperature decreases the rate of the reaction. The temperature at which the enzyme attains its



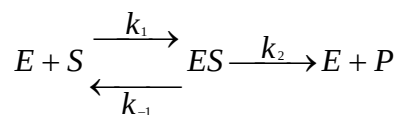
maximum velocity is called optimum temperature. The enhancement of reaction velocity with the increase in temperature from low to optimum is due to: (a) With increase in temperature the initial energy the substrate becomes higher which in turn lowers the activation energy and lowers the energy barrier of the reaction. (b) With the rise in temperature the no of collision between enzyme and substrate increases. Decrease in enzyme activity with the rise in temperature beyond optimum temperature is due to denaturation of the enzyme mainly.

Michaelis- Menten equation

For an enzyme catalyzed reaction, the plot of initial velocity against substrate concentration gives a hyperbolic curve. The nature of the curve can be explained with the help of Michaelis- Menten equation.



It has been assumed that the overall reaction occurs in two steps. In the first step, enzyme (E) interacts with substrate (S) in a reversible manner to form complex (ES). In the next step, the complex dissociates to give enzyme and product (P) irreversibly. This sequence of event is



Now, the initial velocity (rate) of the reaction is

$$v = k_2[ES] \quad \dots\dots\dots (i)$$

[ES] can not be measure experimentally. But [ES] is directly related with total enzyme concentration (E_t) by the following equation

$$[E_t] = [E] + [ES] \quad \dots\dots\dots (ii)$$

$$\text{Rate of ES formation} = k_1[E][S] \quad \dots\dots\dots (iii)$$

$$= k_1([E_t] - [ES])[S] \quad \dots\dots\dots (iv)$$

$$\text{Rate of ES breakdown} = (k_{-1} + k_2)[ES] \quad \dots\dots\dots (v)$$

Applying the rule of steady state approximation for ES, we have,

$$\text{Rate of ES formation} = \text{Rate of ES breakdown}$$

$$k_1([E_t] - [ES])[S] = (k_{-1} + k_2)[ES]$$

$$[ES](k_{-1} + k_2 + k_1[S]) = k_1[E_t][S]$$

$$[ES] = \frac{k_1[E_t][S]}{k_1[S] + (k_2 + k_{-1})} \quad \dots\dots\dots (vi)$$

Dividing the numerator and denominator by k_1

$$[ES] = \frac{[E_t][S]}{[S] + \left(\frac{k_2 + k_{-1}}{k_1}\right)} \quad \dots\dots\dots (vii)$$

Now defining, $\left(\frac{k_2 + k_{-1}}{k_1}\right) = K_m$, known as Micheelis- Menten constant, we have,

$$[ES] = \frac{[E_t][S]}{[S] + K_m} \quad \dots\dots\dots (viii)$$

So, the initial velocity (rate) of the reaction is

$$v = k_2[ES] \quad \dots\dots\dots (ix)$$

$$v = \frac{k_2[E_t][S]}{[S] + K_m} \quad \dots\dots\dots (x)$$

At high substrate concentration, all enzyme is saturated with substrate i.e. all enzyme in the ES form and the velocity of the reaction attains its maximum value.

Mathematically,

$$\text{When } [E_t] = [ES], \text{ then } v = V_{\max}$$

Putting these relations in equation (ix) we have,

$$V_{\max} = k_2[E_t] \quad \dots\dots\dots (xi)$$

Combining eqn. (x) and (xi), we obtain

$$v = \frac{V_{\max}[S]}{[S] + K_m} \quad \dots\dots\dots (xii)$$

This eqn. is known as Michaelis- Menten equation

If the initial velocity, $v = \frac{1}{2}V_{\max}$, applying in eqn. (xii) we have

$$\frac{1}{2}V_{\max} = \frac{V_{\max}[S]}{[S] + K_m}$$

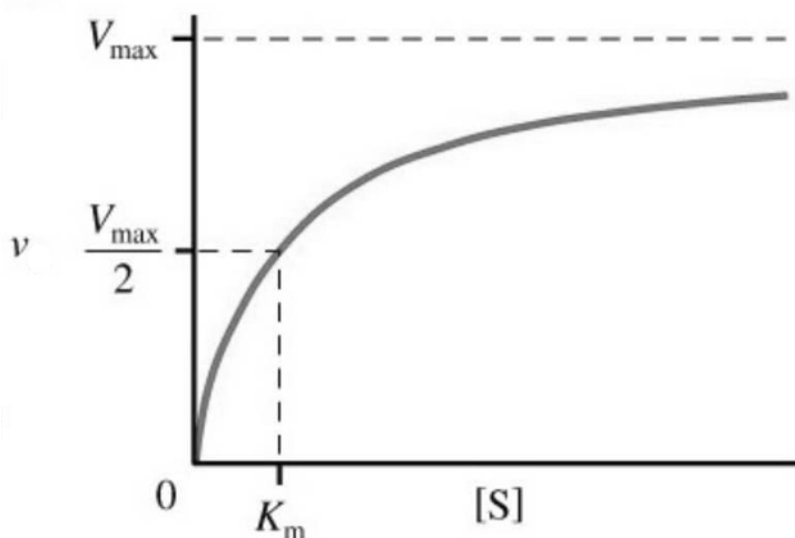
Simplifying, $K_m = [S]$

Hence, K_m is the substrate concentration when the velocity of the reaction becomes half of its maximum value.

Small K_m value for a enzyme indicates that the reaction acquires maximum catalytic efficiency at low substrate concentration.

Small K_m value indicates tight binding between enzyme and substrate whereas high value of K_m indicates weak binding.

Unit of K_m : It has the same unit as that of the substrate concentration.



The Lineweaver-Burk plot: Determination of K_m and V_{max}

The direct plot of initial velocity (v) vs substrate concentration ($[S]$) using Micheelis- Menten equation does not give accurate measurement of V_{max} and hence K_m . In this curve, the initial velocity (v) approaches V_{max} asymptotically at very high substrate concentration. But even at 10 time's greater substrate concentration than K_m , the initial velocity (v) is only 91% of that V_{max} . So the value of V_{max} , obtained from the extrapolation of the asymptote will not be accurate. To overcome this problem, Hans Lineweaver and Dean Burk use the reciprocal of eqn. (xii).

$$v = \frac{V_{max}[S]}{[S] + K_m}$$

$$\frac{1}{v} = \frac{[S] + K_m}{V_{max}[S]}$$

$$\frac{1}{v} = \frac{K_m}{V_{max}[S]} + \frac{[S]}{V_{max}[S]}$$

$$\frac{1}{v} = \frac{K_m}{V_{max}} \frac{1}{[S]} + \frac{1}{V_{max}}$$

The plot of $1/v$ (X- axis) against $1/[S]$ (Y-axis) generates a straight line with slope = K_m/V_{max} . This plot is known as Lineweaver- Burk plot or double reciprocal plot. Extrapolation of the straight line gives intercepts $1/V_{max}$ in the Y-axis and $-1/K_m$ in the X-axis from which V_{max} and K_m can be determine easily.

